Neural Amortized Inference for Nested Multi-agent Reasoning

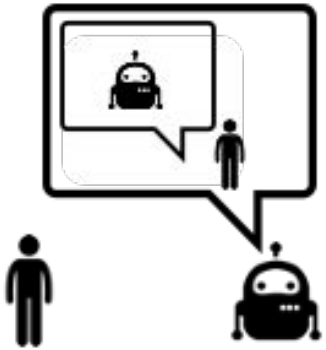
Kunal Jha, Tuan Anh Le, Chuanyang Jin, Yen-Ling Kuo, Joshua Tenenbaum and Tianmin Shu



DARTMOUTH



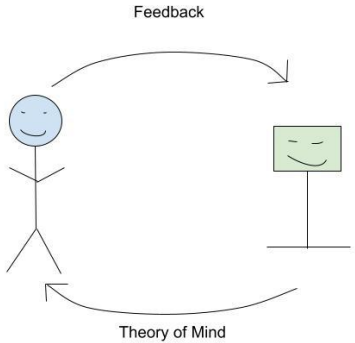
(July 2023)



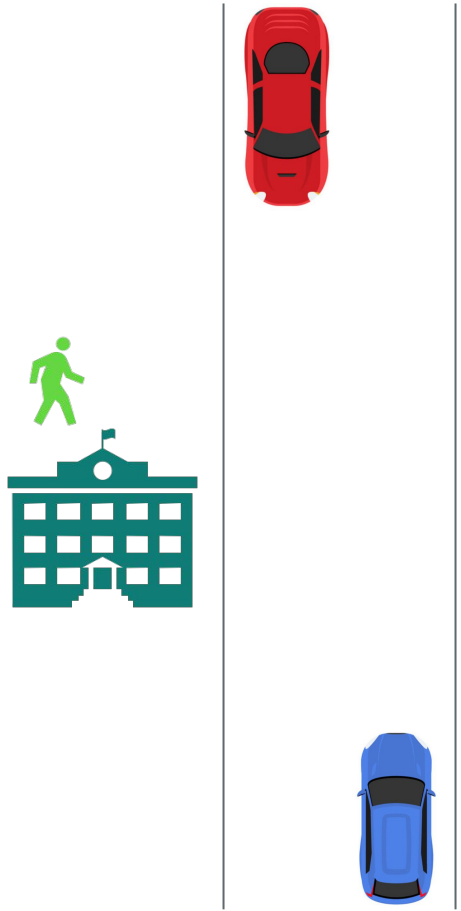
Theory of Mind

- Theory of Mind: Reasoning about the hidden mental variables that drive observable actions

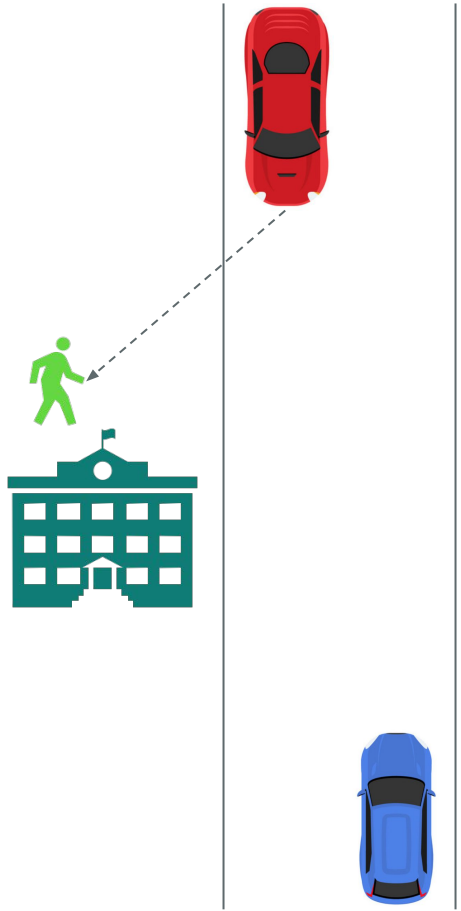




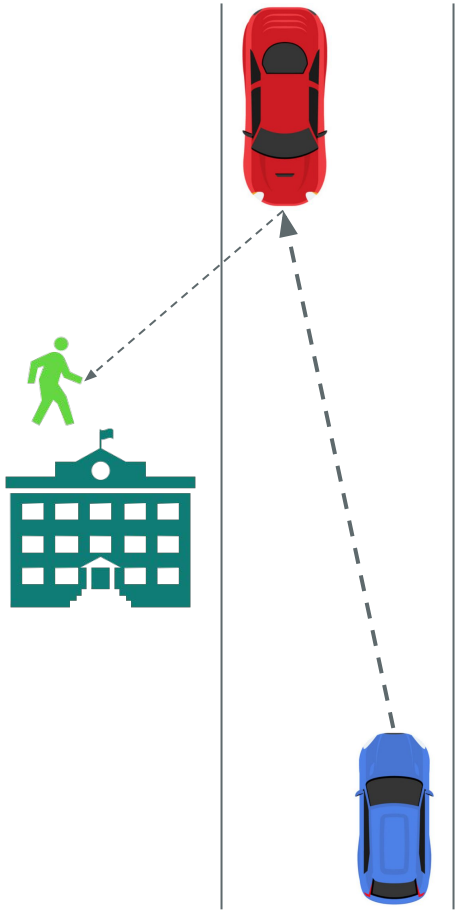
The blue car sees the red car but can’t see the pedestrian behind the building.



The red car sees the pedestrian
and slows down to avoid crashing.



The blue car sees the red car slowing down. Not seeing any other obstacles, it infers the presence of a pedestrian behind the building and therefore slows down.



How Can We Model This?

Partially Observable Markov Decision Processes (POMDP)



- The entire state of the world is not visible
- Each agent stores (*mental states*, *actions*, *observations*)



Bayesian Inverse Planning

P(mental state | plan) $\propto$ P(plan | mental state) * P(mental state)

Interactive Partially Observable Markov Decision Processes (I-POMDP)

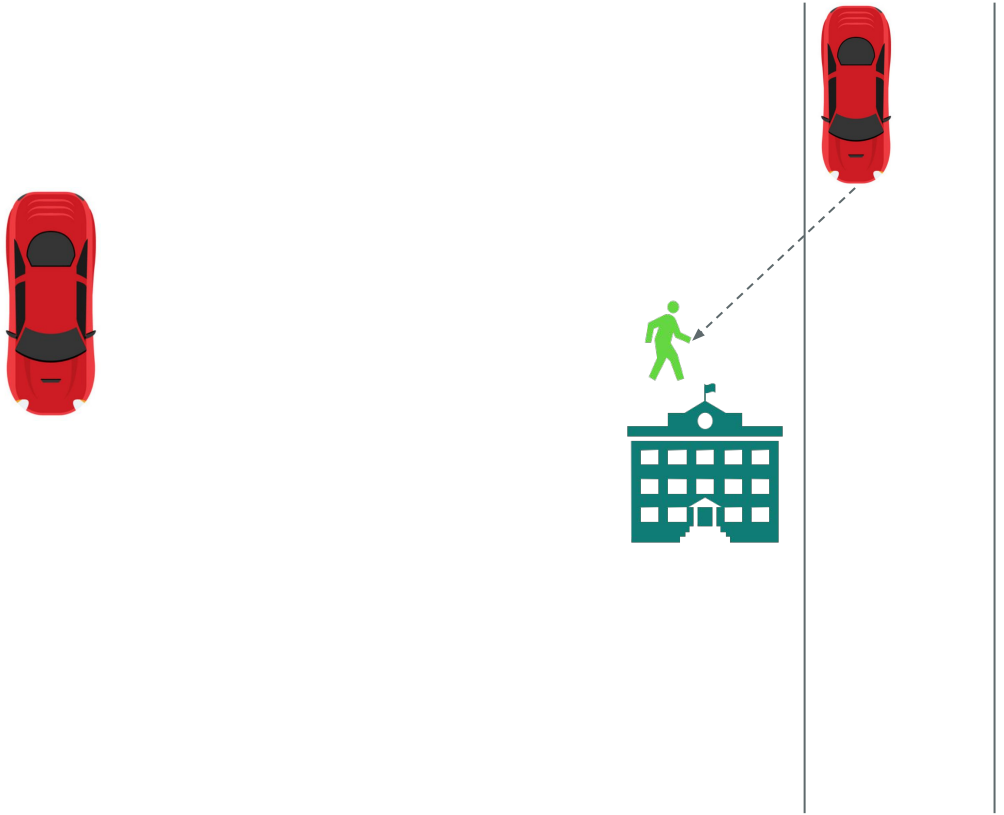




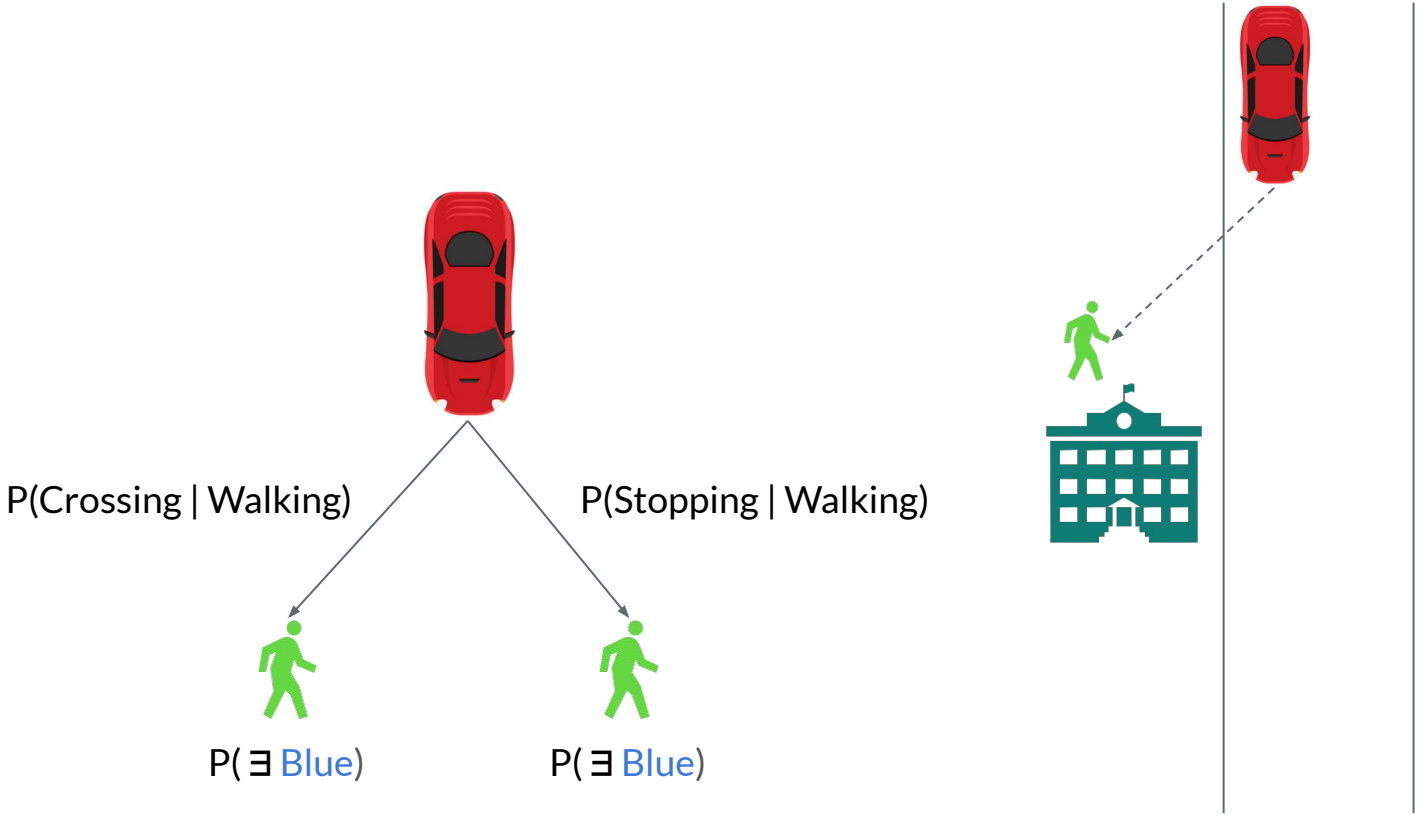


P(Blue car existing behind the barrier) ≔ P(∃ Blue)

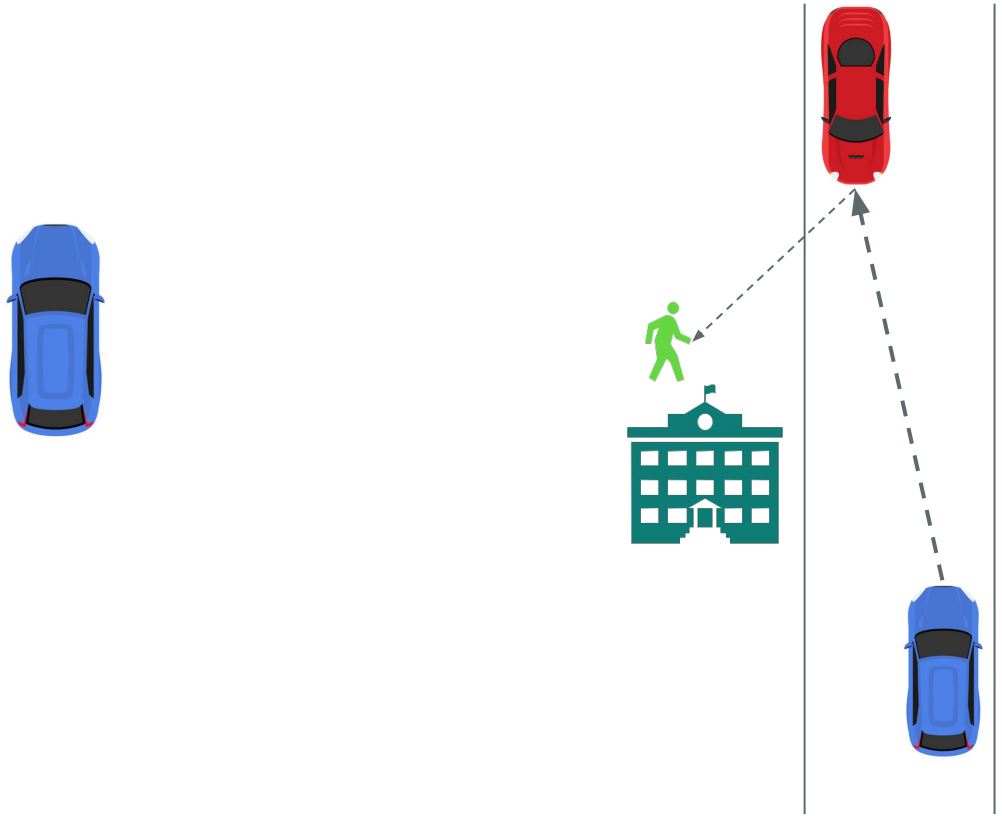
Interactive Partially Observable Markov Decision Processes (I-POMDP)



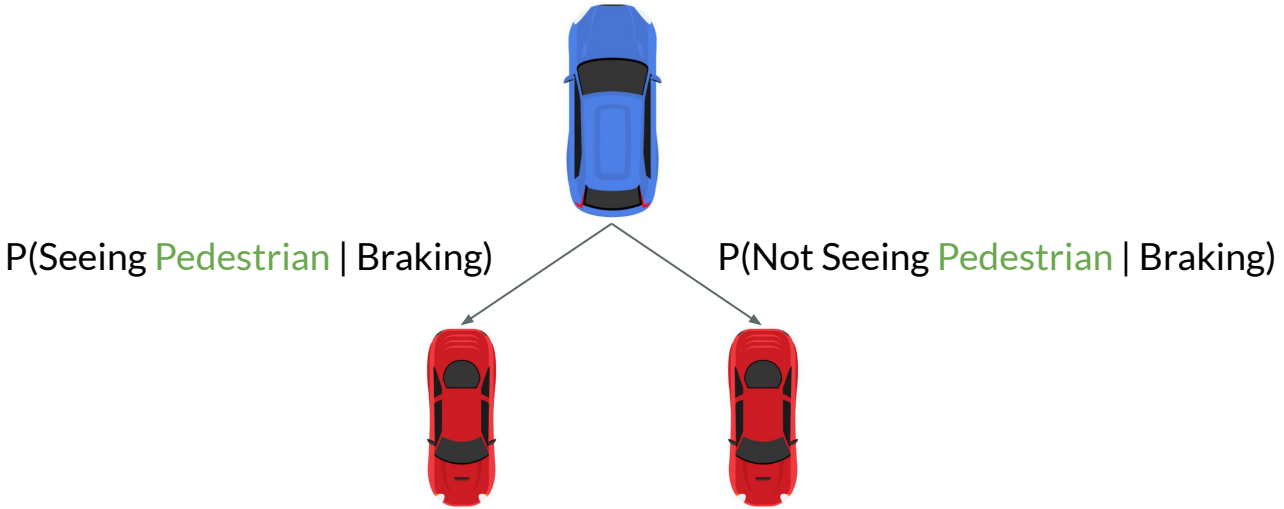
Interactive Partially Observable Markov Decision Processes (I-POMDP)

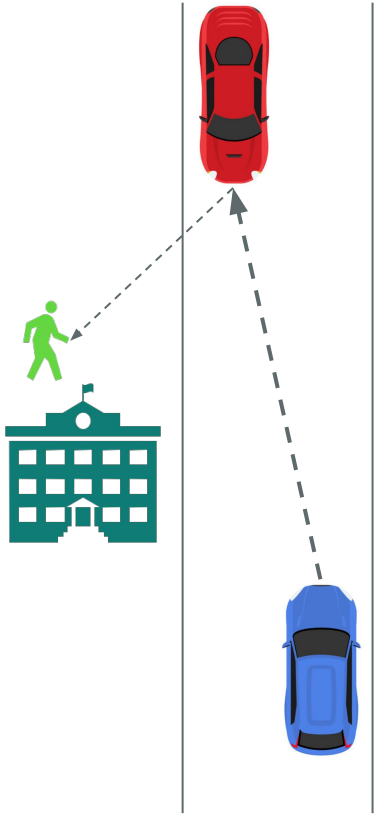


Interactive Partially Observable Markov Decision Processes (I-POMDP)

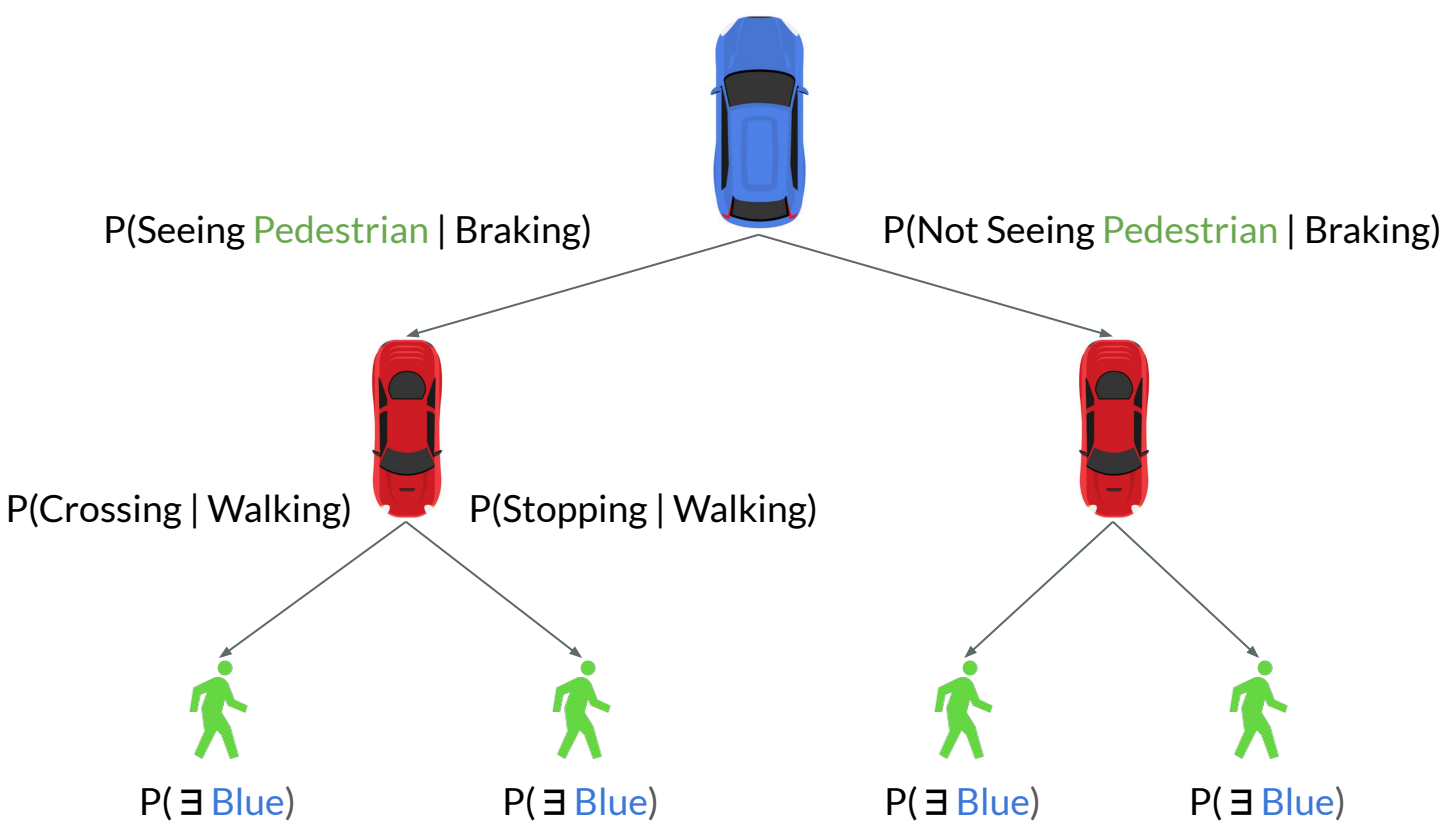


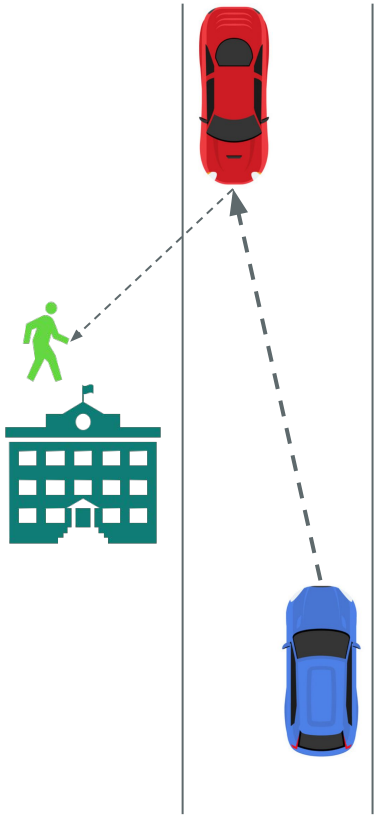
Interactive Partially Observable Markov Decision Processes (I-POMDP)





Interactive Partially Observable Markov Decision Processes (I-POMDP)





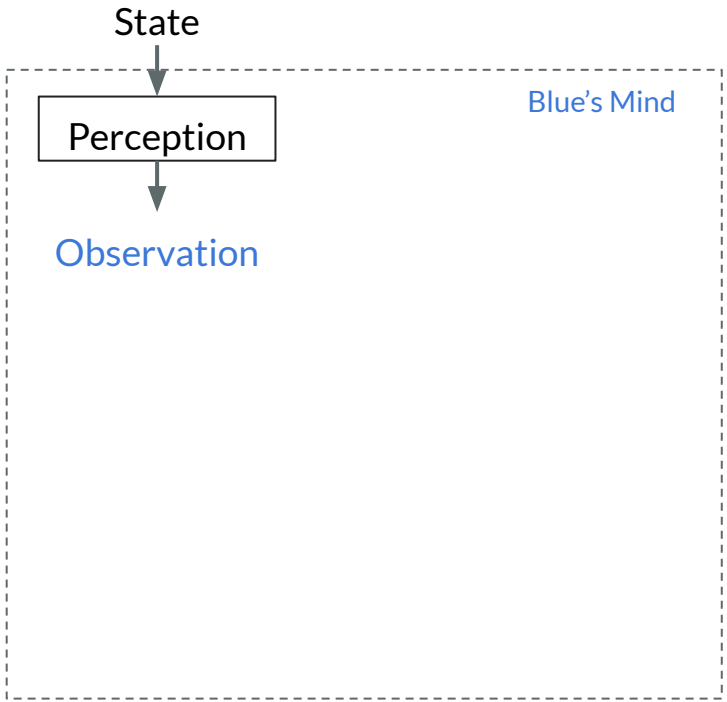
Background - Interactive Partially Observable Markov Decision Processes

State

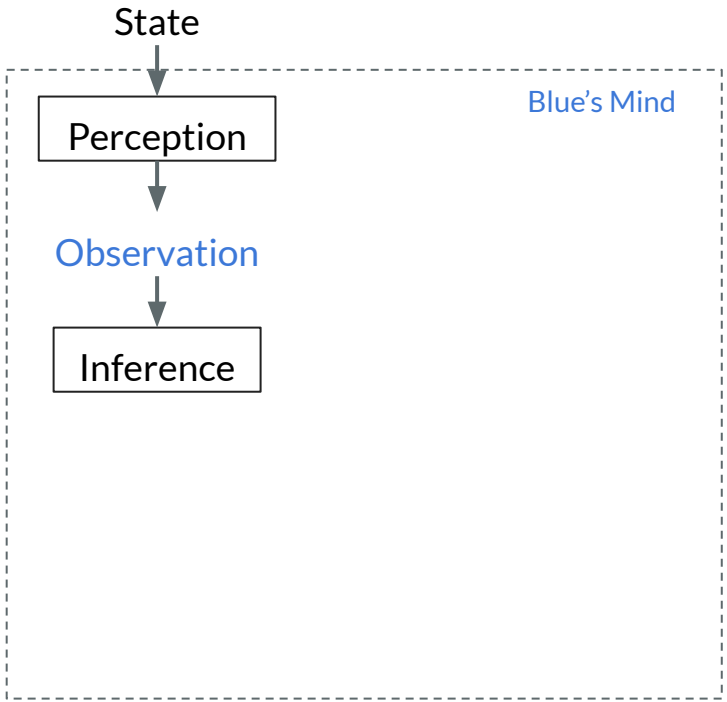
Background - Interactive Partially Observable Markov Decision Processes



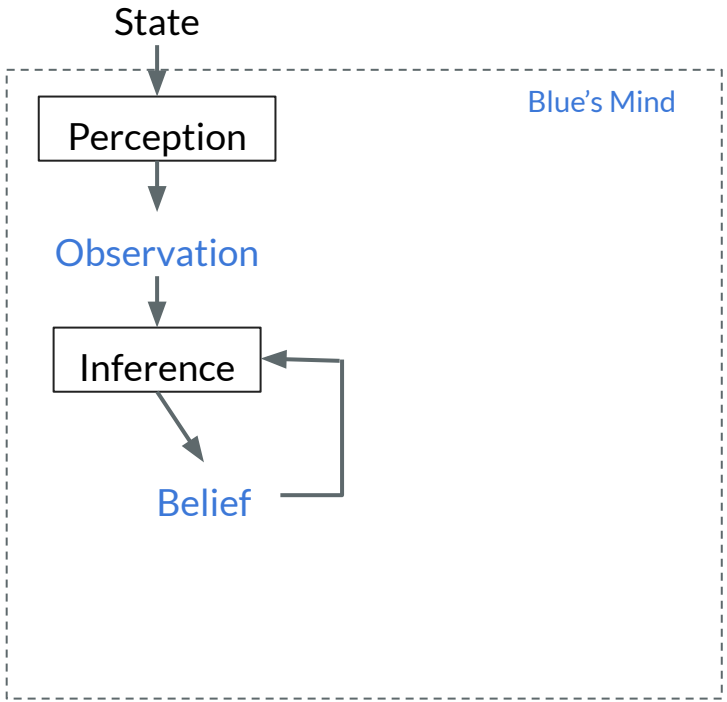
Background - Interactive Partially Observable Markov Decision Processes



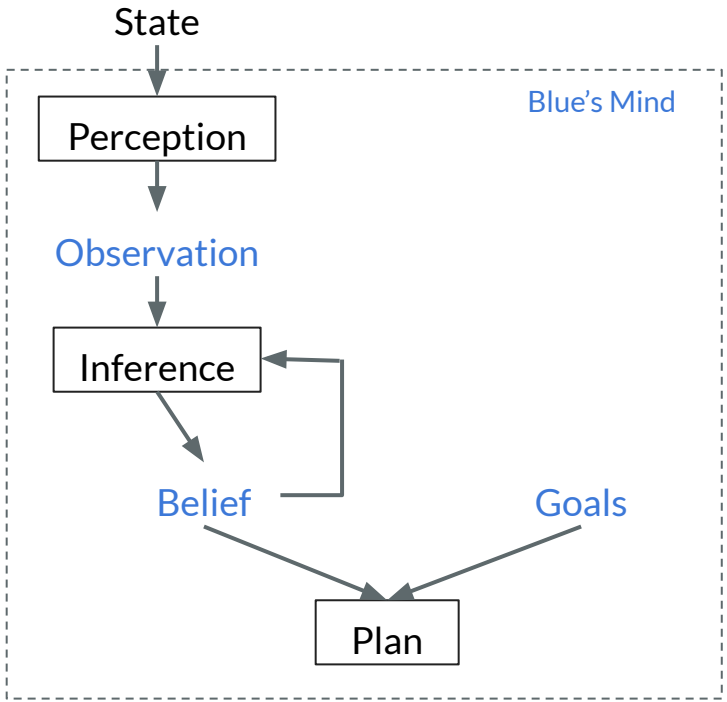
Background - Interactive Partially Observable Markov Decision Processes



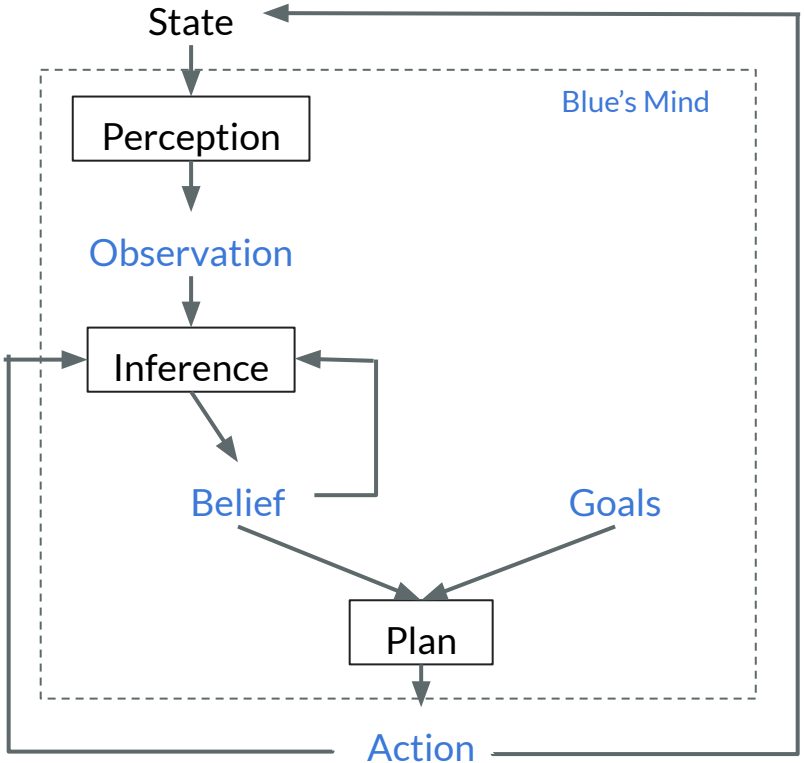
Background - Interactive Partially Observable Markov Decision Processes



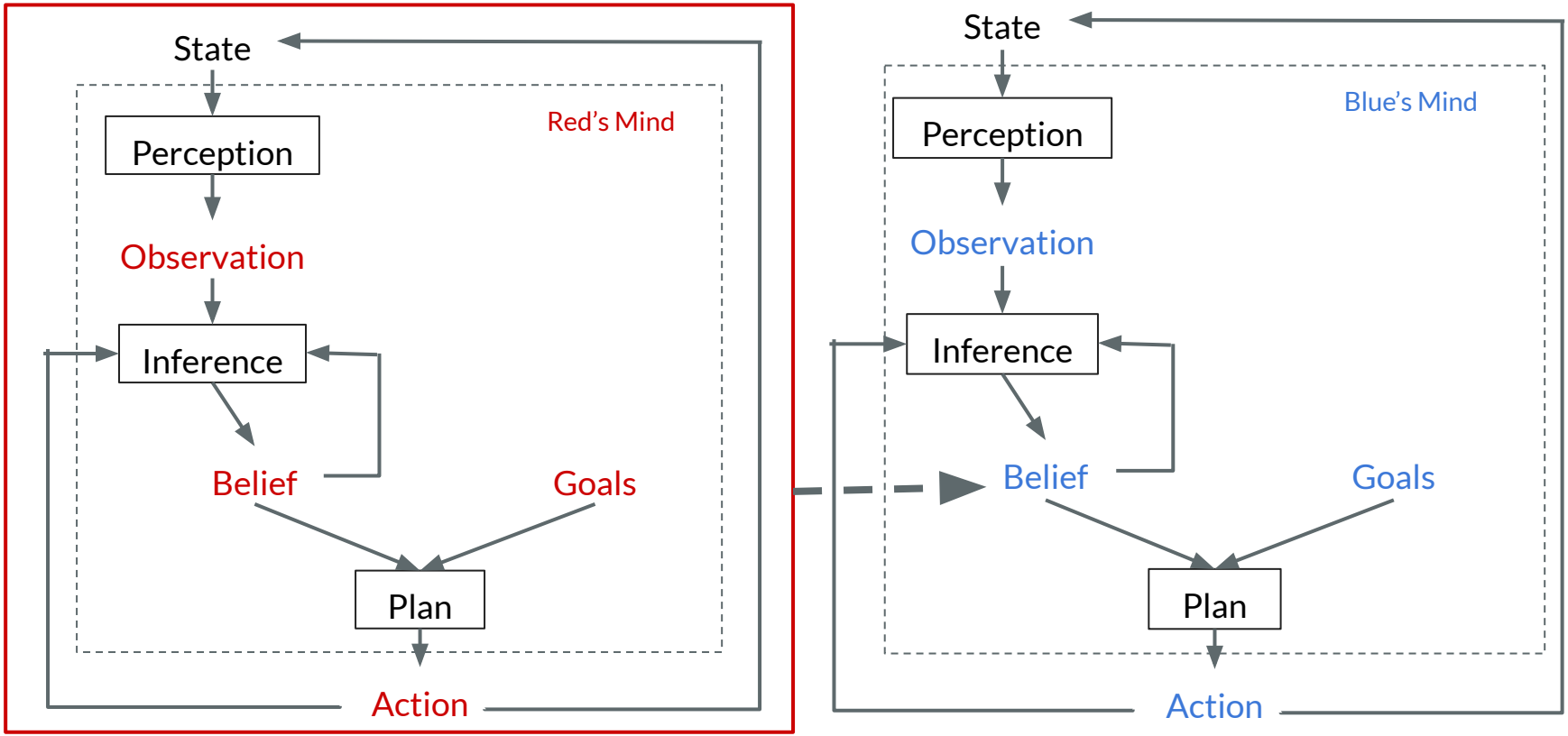
Background - Interactive Partially Observable Markov Decision Processes



Background - Interactive Partially Observable Markov Decision Processes

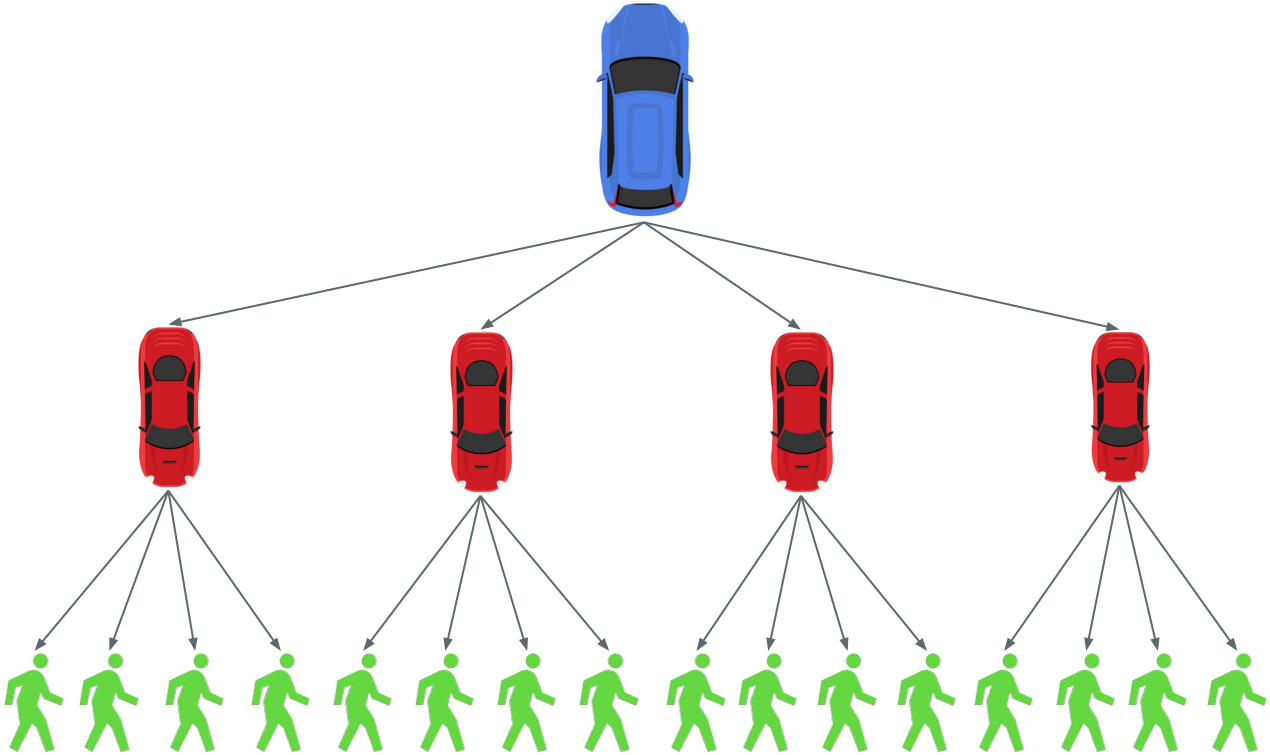


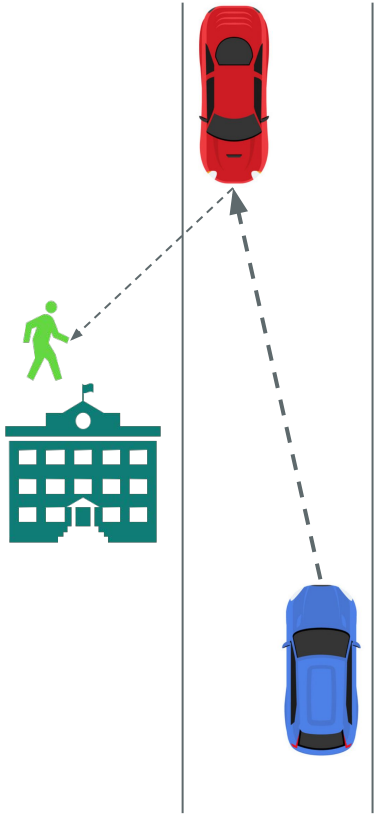
Background - Interactive Partially Observable Markov Decision Processes



Our Approach: Amortize Lower Levels and Update Beliefs through Importance Sampling

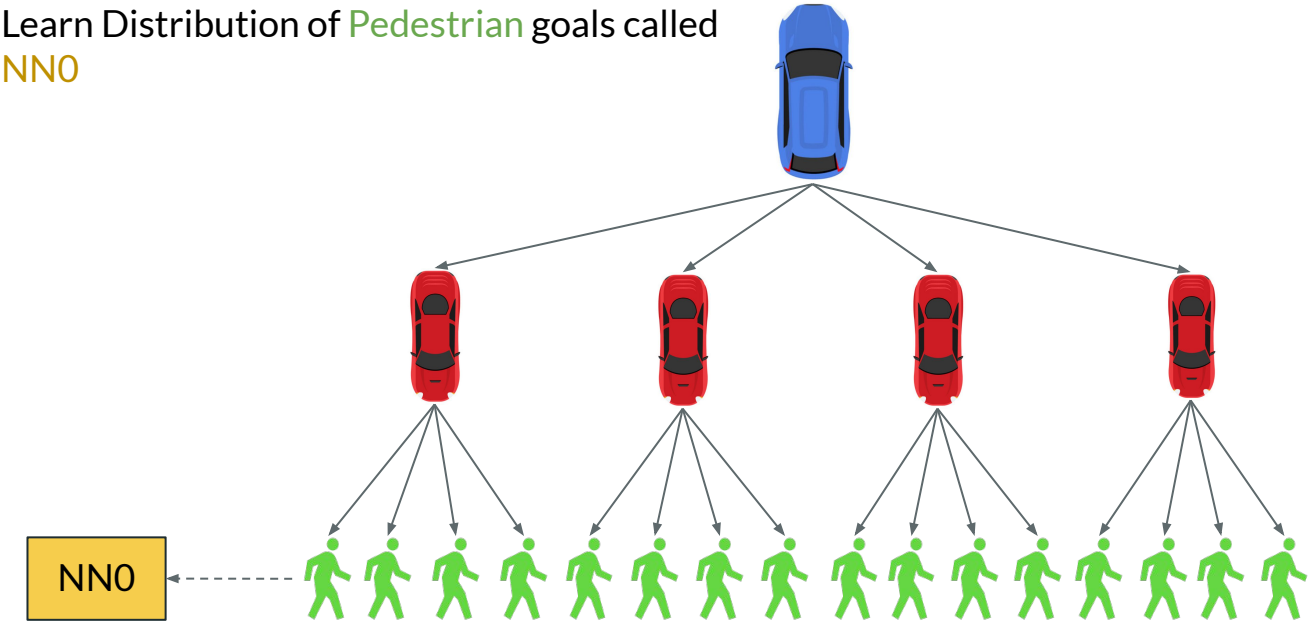
Our Approach – Amortized Inference

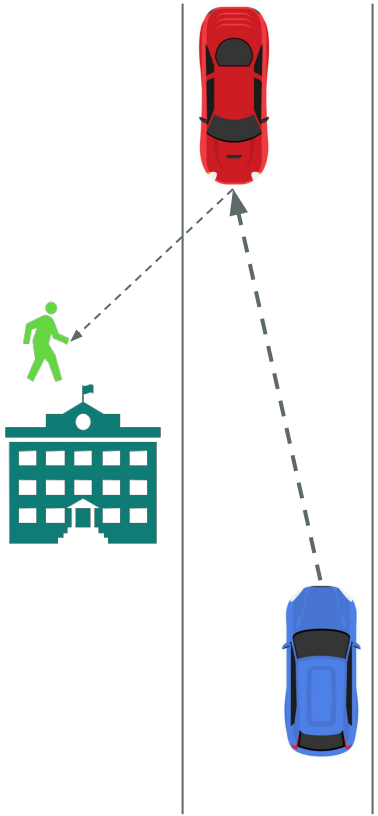




Our Approach – Amortized Inference

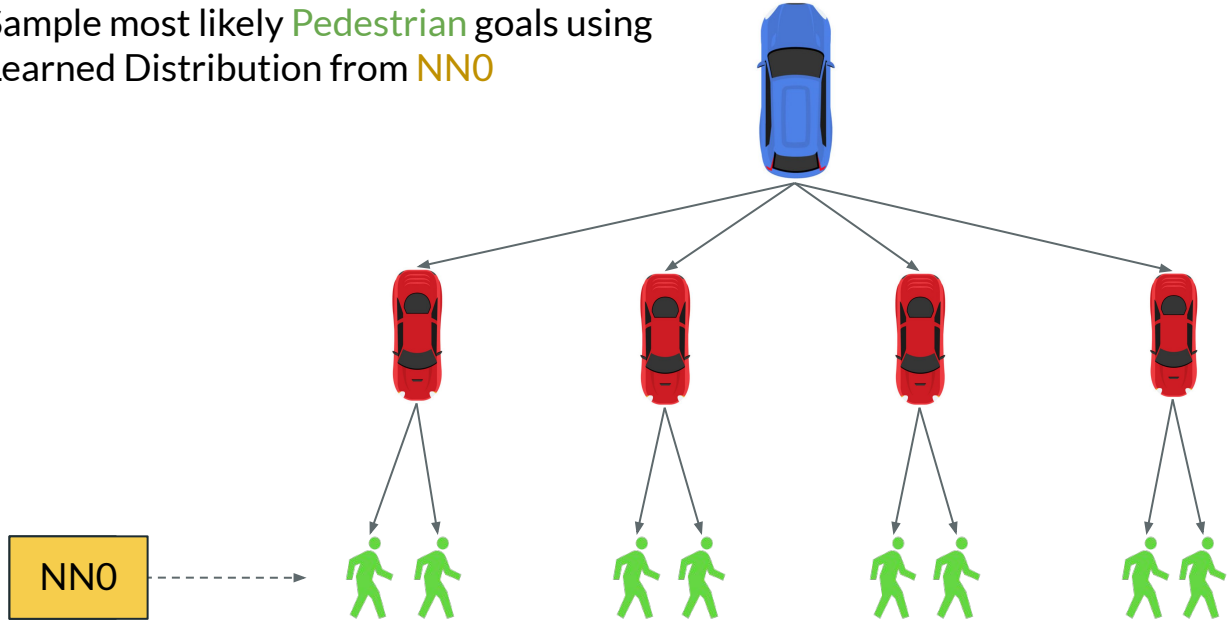
Learn Distribution of Pedestrian goals called NN0

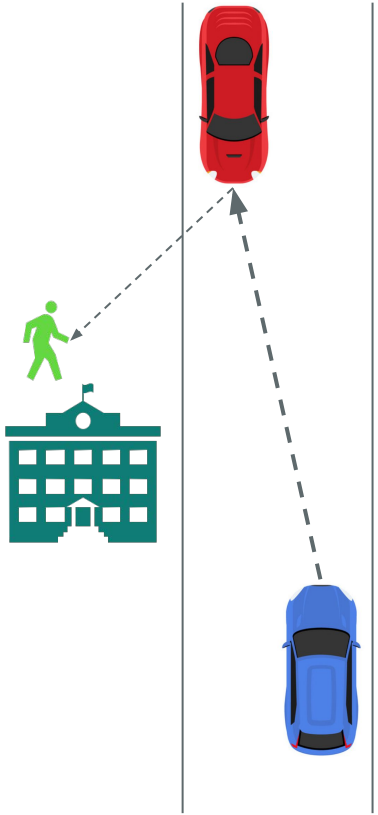




Our Approach – Amortized Inference

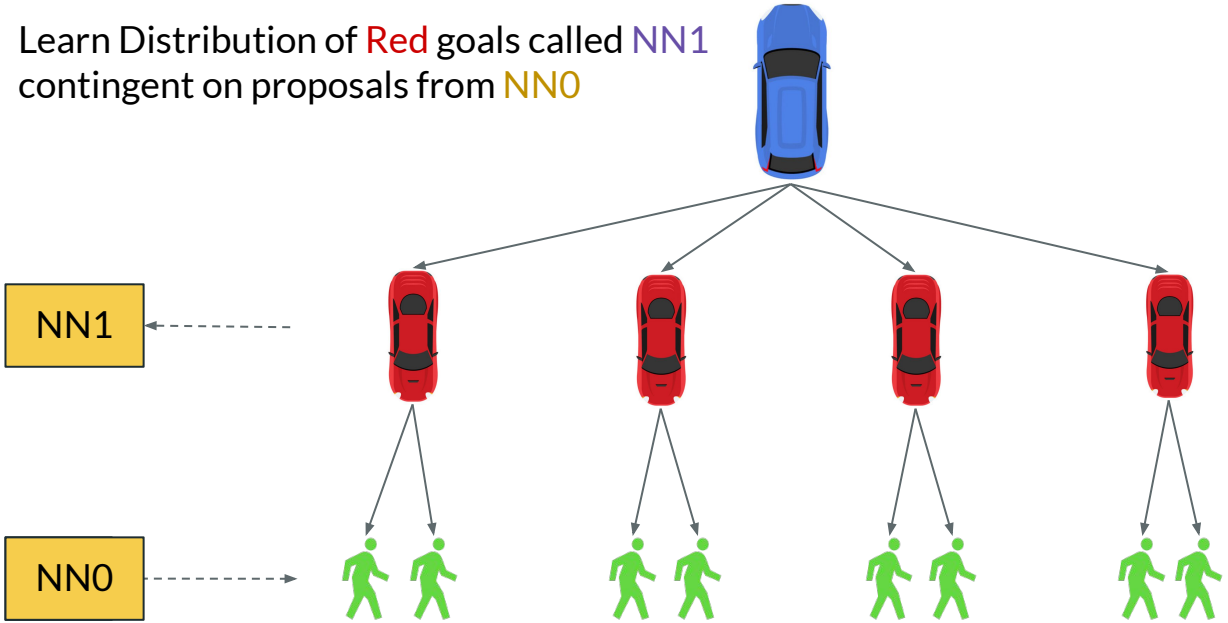
Sample most likely Pedestrian goals using Learned Distribution from NN0

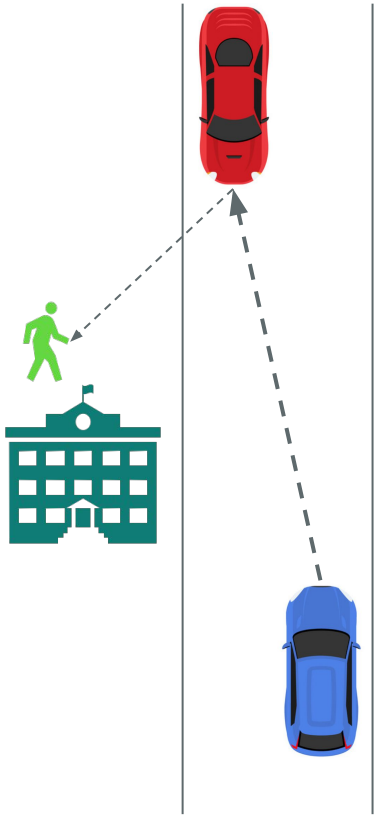




Our Approach – Amortized Inference

Learn Distribution of Red goals called NN1
contingent on proposals from NN0



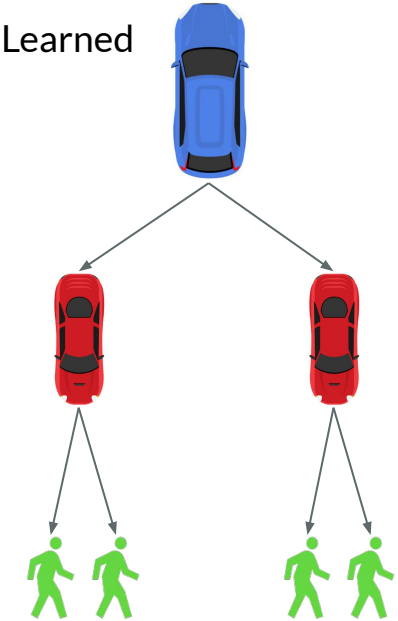


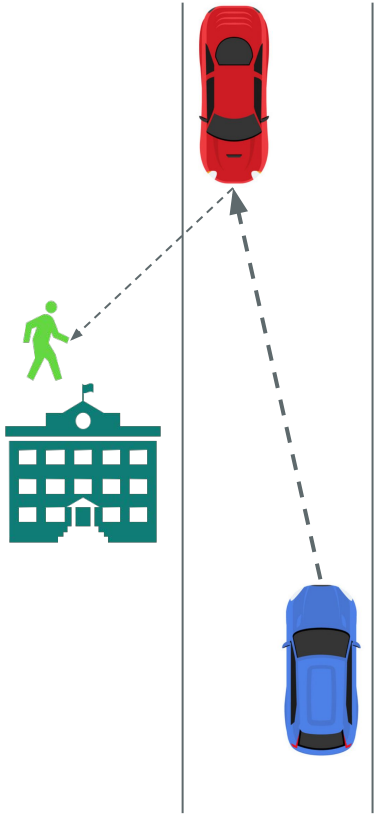
Our Approach – Amortized Inference

Sample most likely Red goals using Learned Distribution from NN1

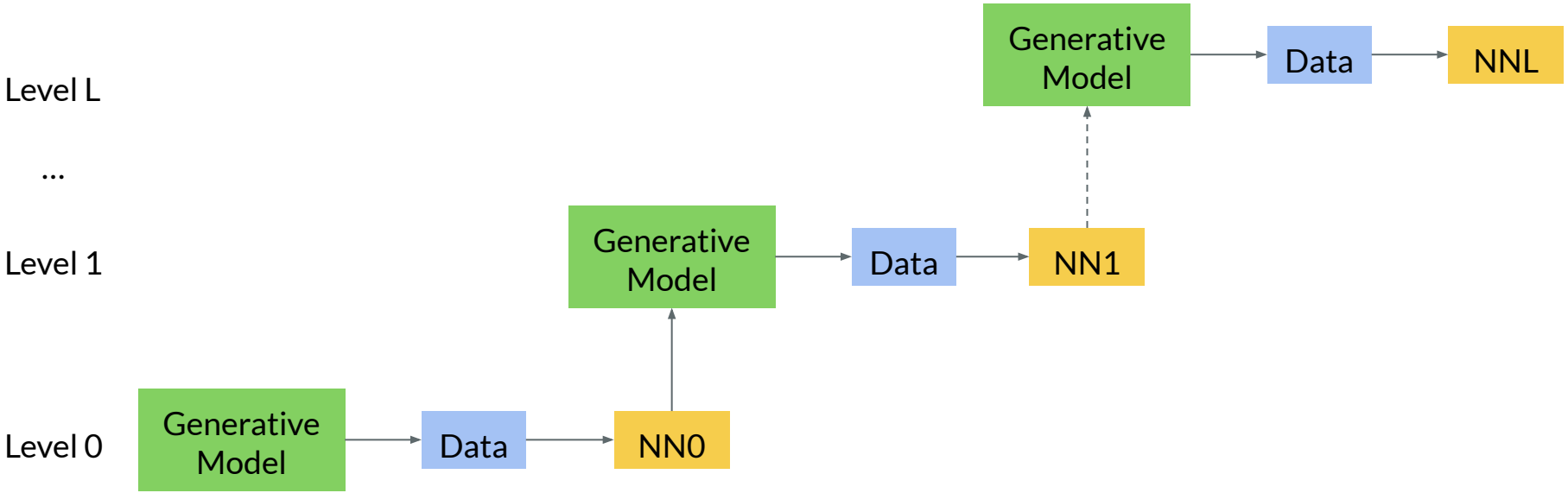
NN1

NN0





Our Approach – Building a Dataset for Recursive Reasoning



Experiment - Setup

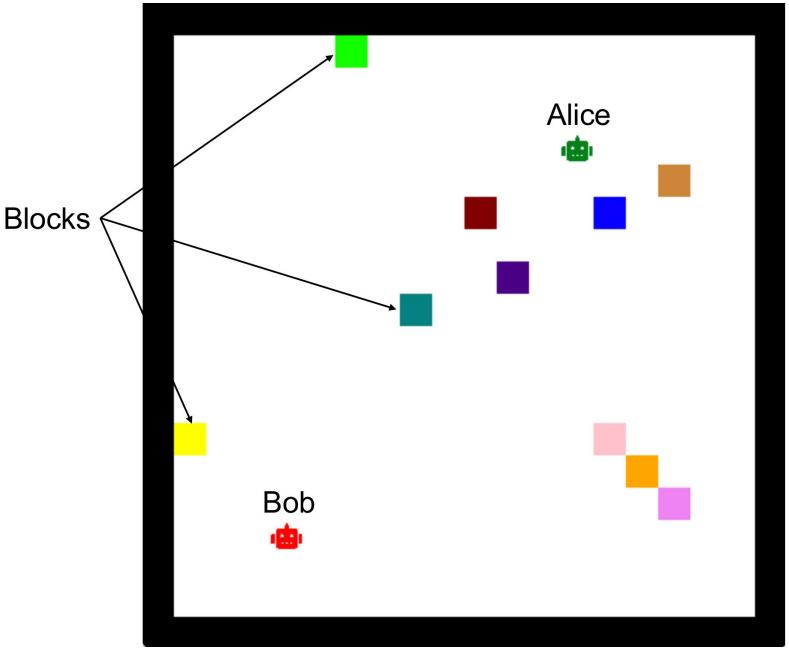
- 2 Agents, Alice (Green), Bob (Red)
  - Alice wants to move two colored blocks together
  - Bob wants to help or hinder Alice
- Neither agent knows each other's goal
- At each timestep, we ask a model to infer the intentions of Bob
- Goal Space **20 times larger** than prior work



Alice

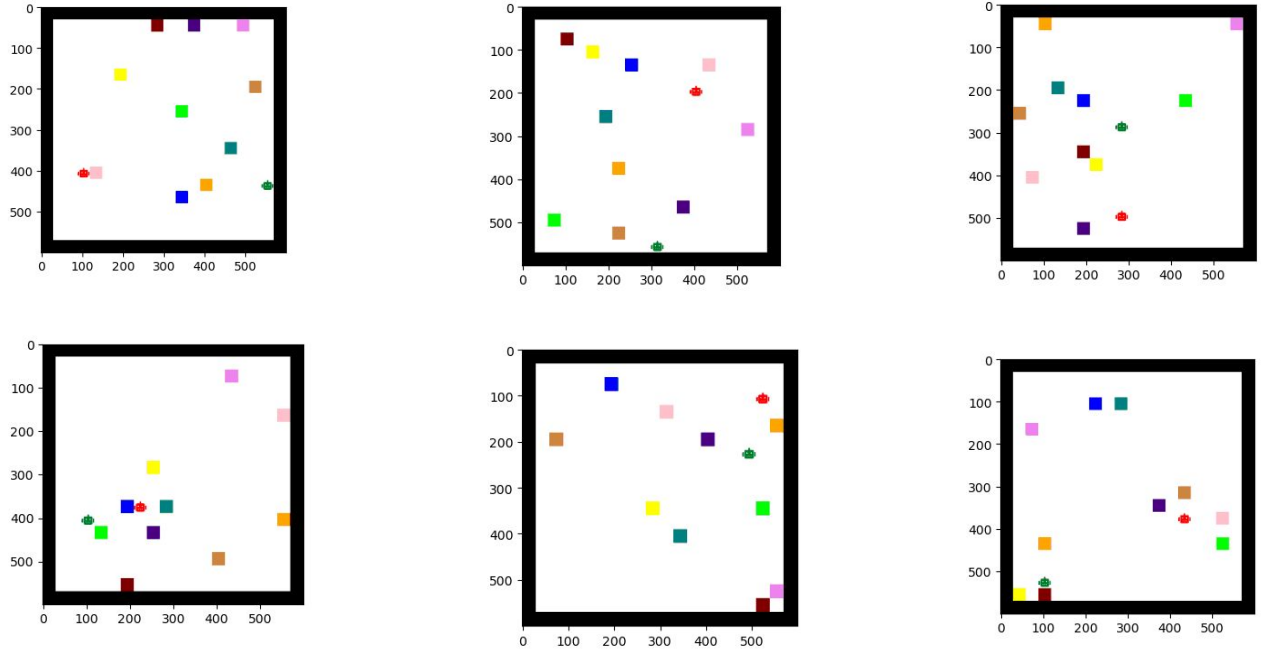


Bob

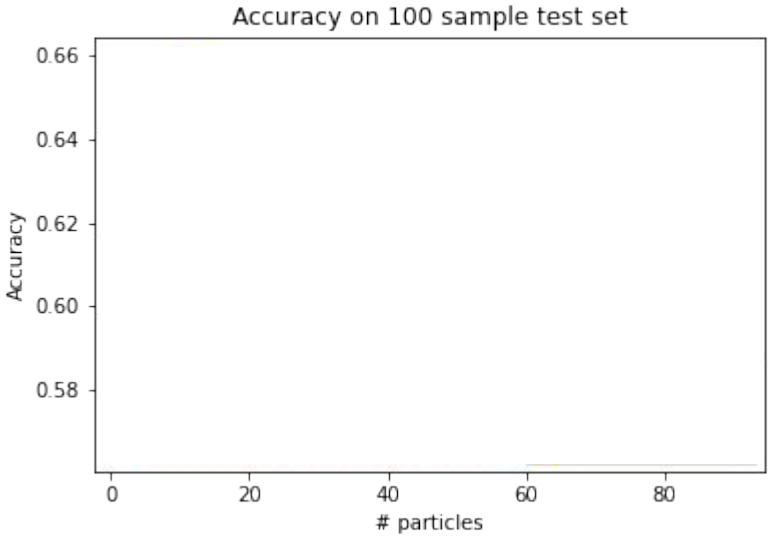


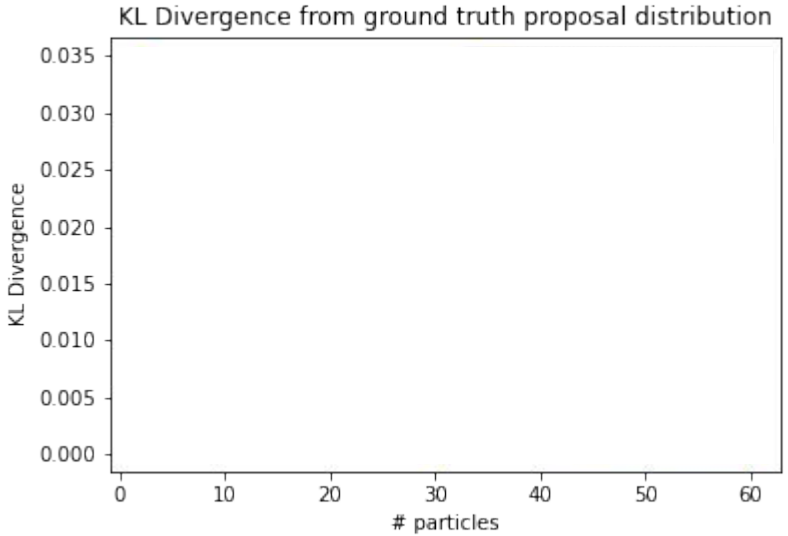
Experiment - Data Generation

- Procedurally Generate Data by sampling *random States and Goals*

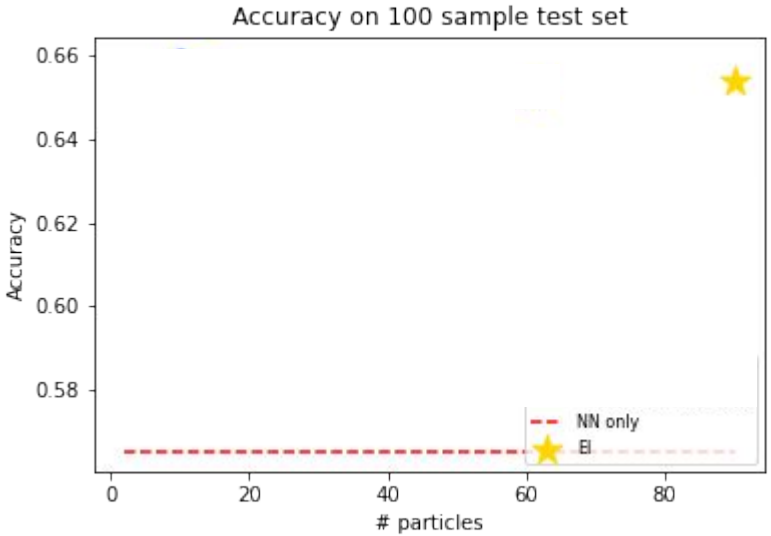


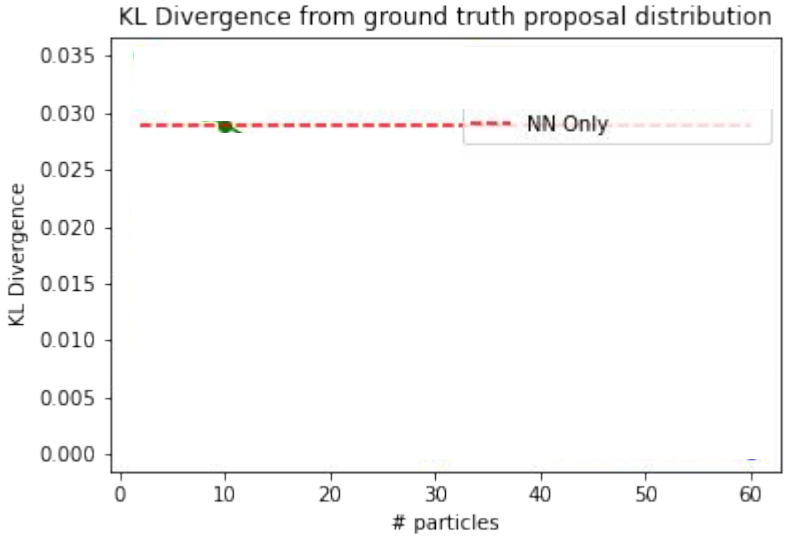
Experiment - Results on Larger Dataset



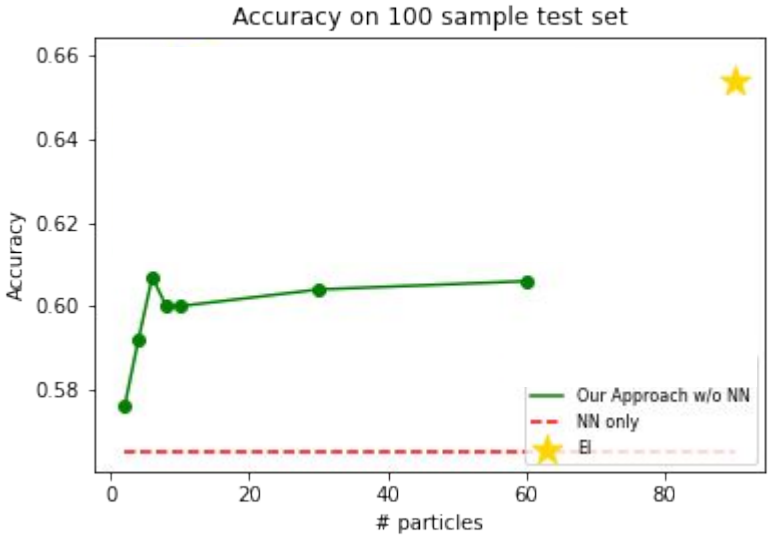


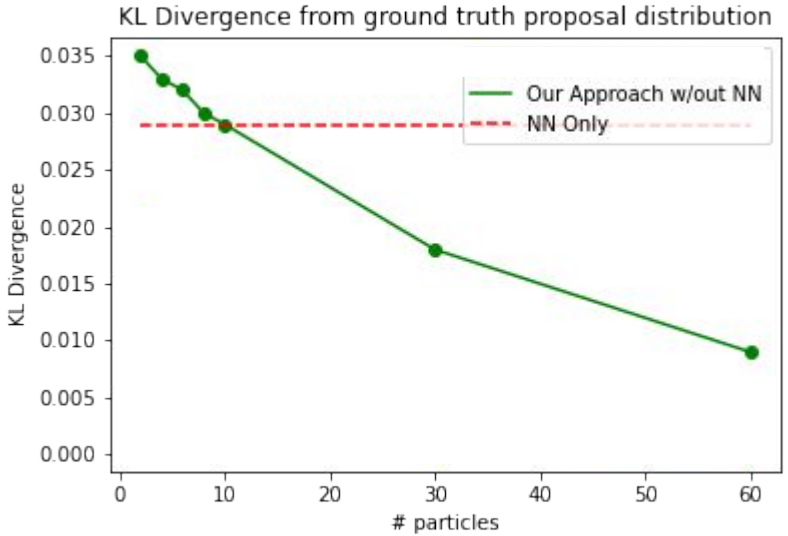
Experiment - Results on Larger Dataset



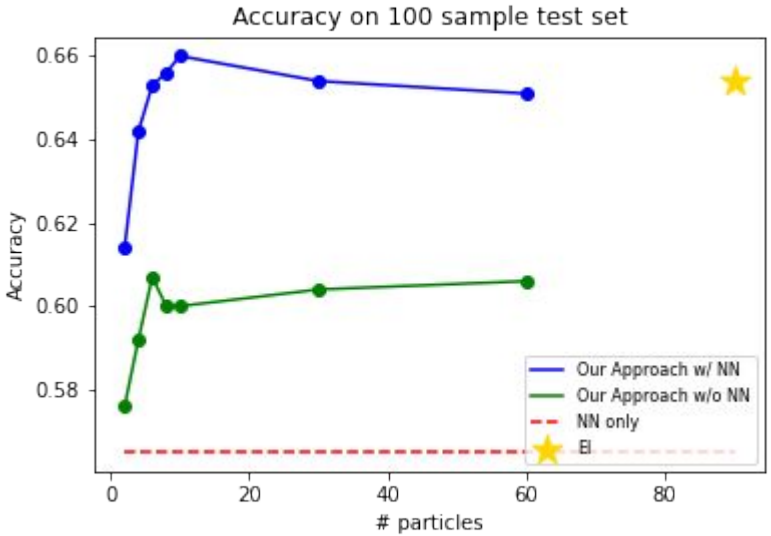


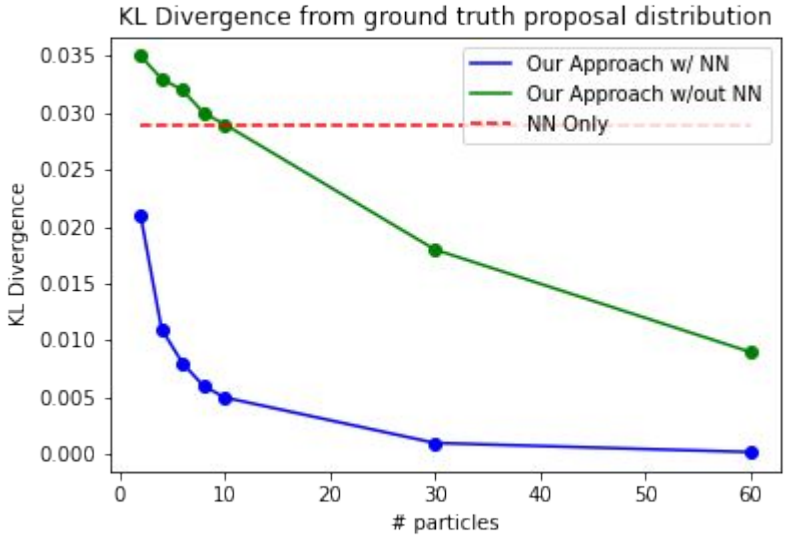
Experiment - Results on Larger Dataset





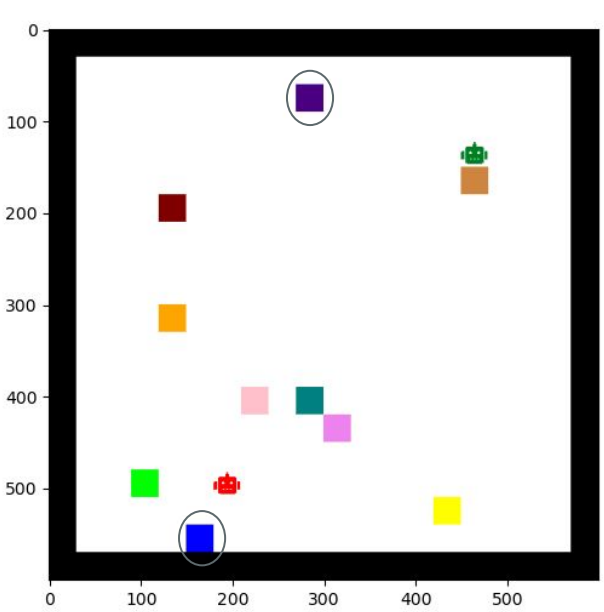
Experiment - Results on Larger Dataset

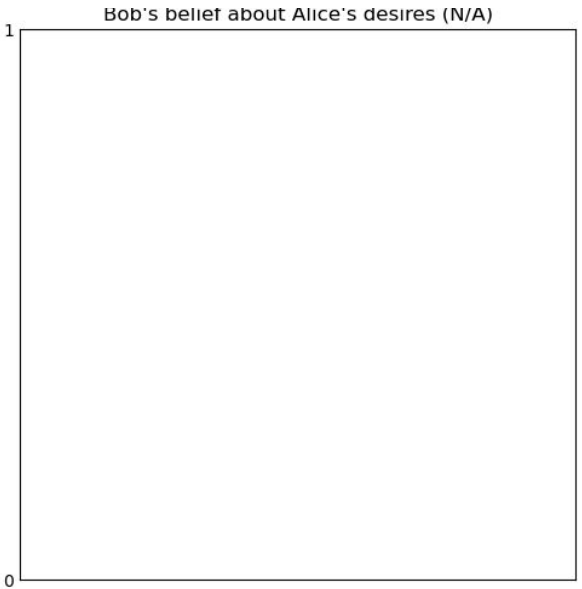


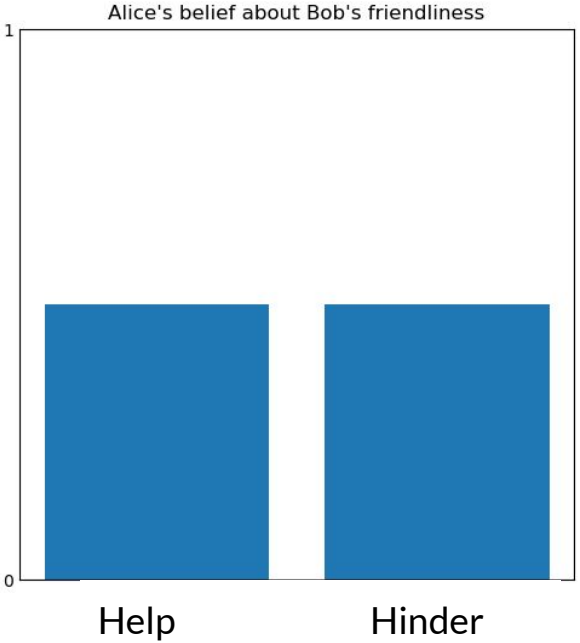


Experiment – Results from Bob Helping Alice

Alice's Goals: and



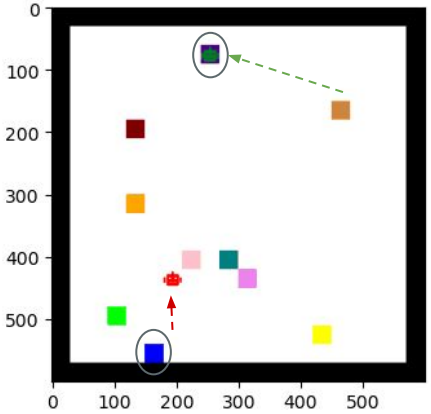


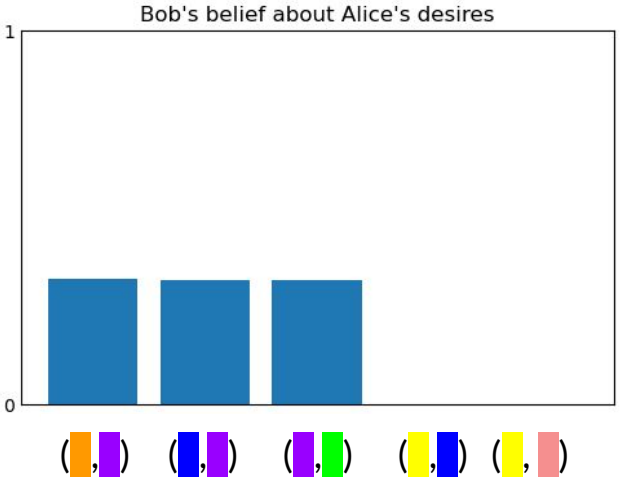


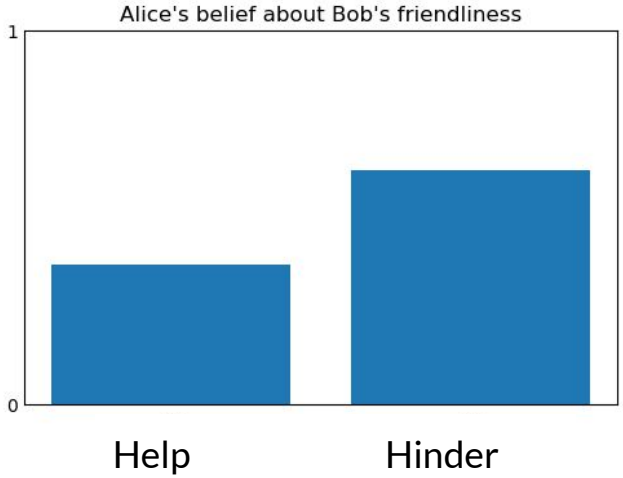
Experiment - Results from Bob Helping Alice

Alice's Goals: ■ and ■

Bob's Social Goal: Help



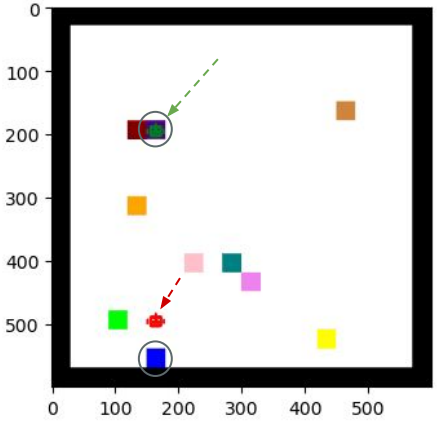


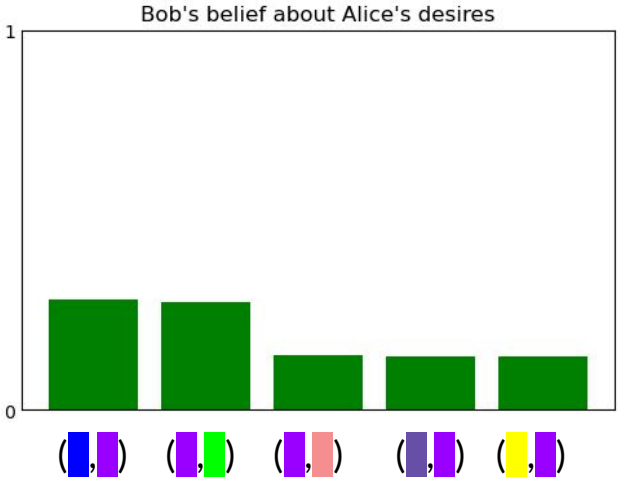


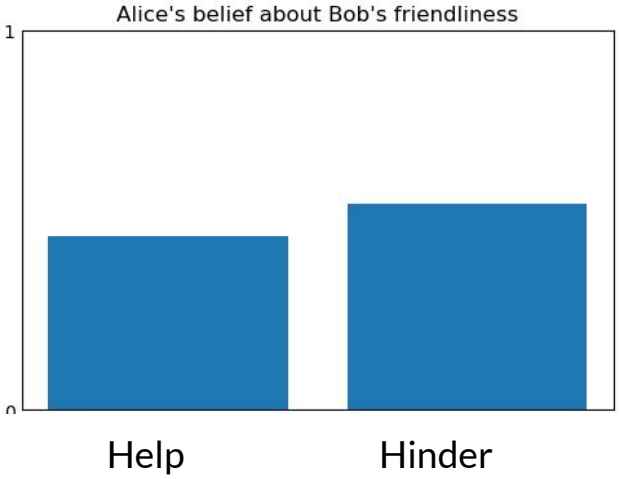
Experiment - Results from Bob Helping Alice

Alice's Goals: ■ and ■

Bob's Social Goal: Help

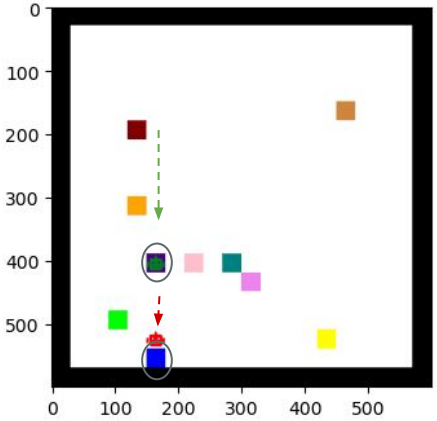


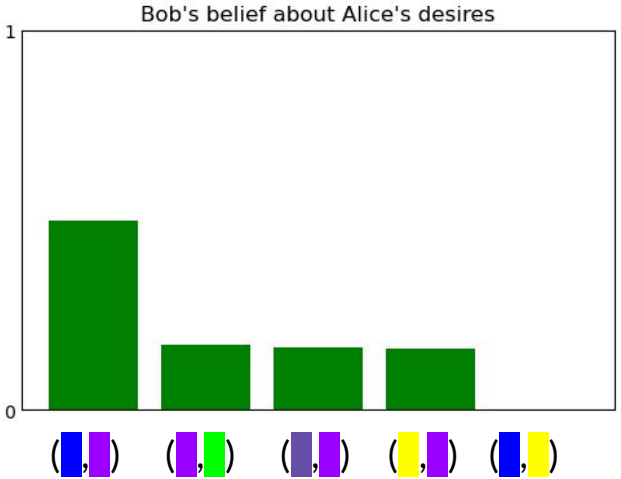


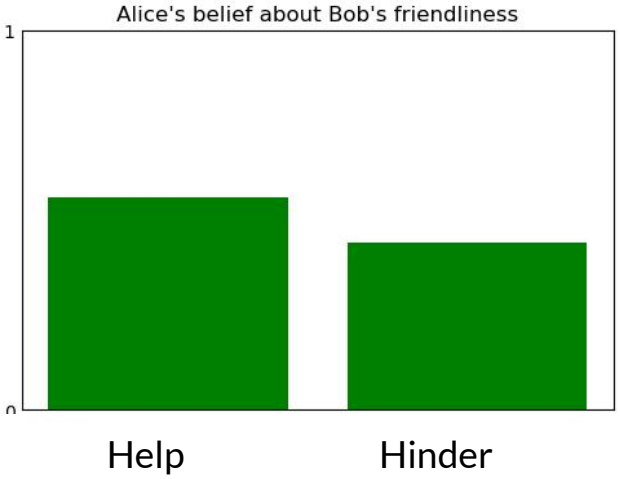


Experiment - Results from Bob Helping Alice

Alice's Goals: ■ and ■
Bob's Social Goal: Help



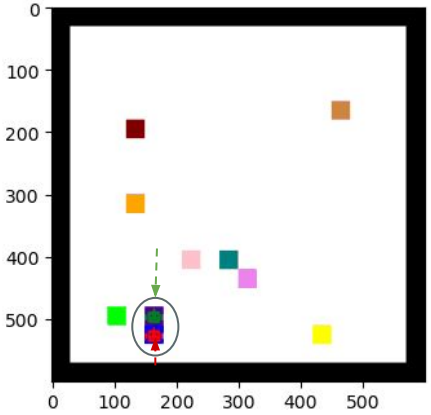


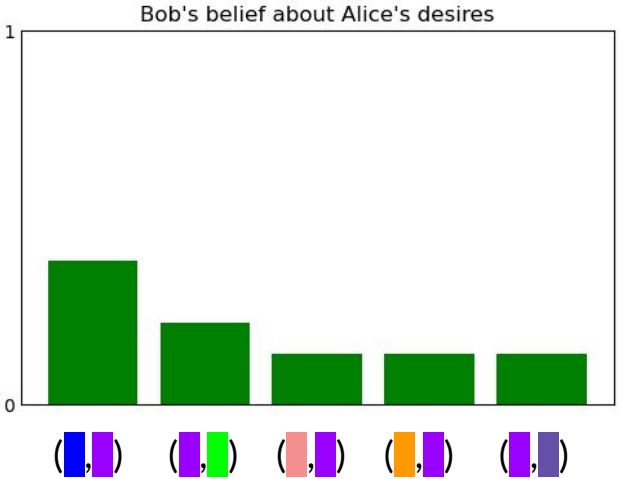


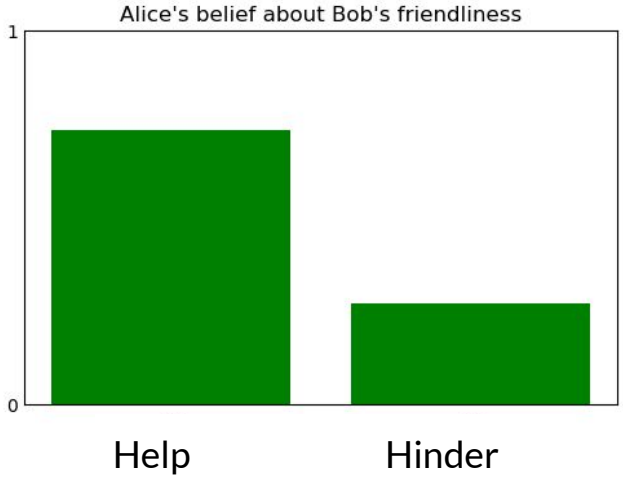
Experiment - Results from Bob Helping Alice

Alice's Goals: ■ and ■

Bob's Social Goal: Help

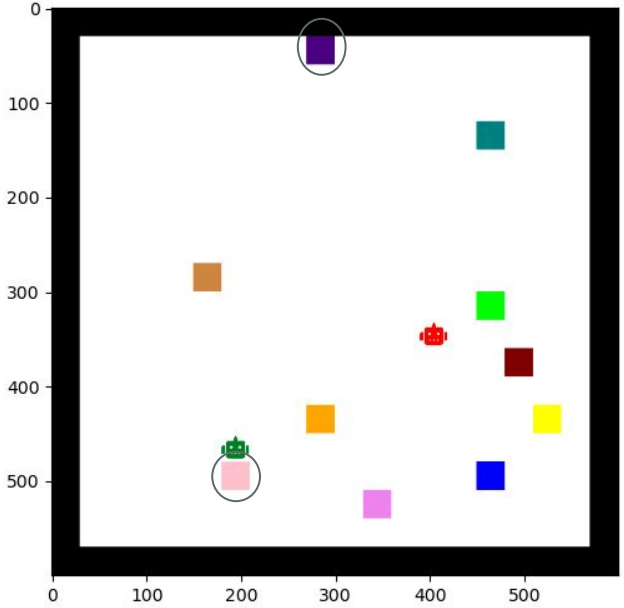


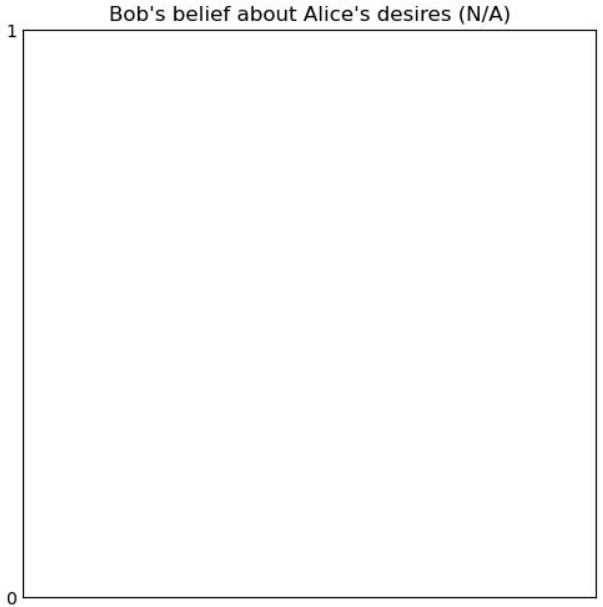


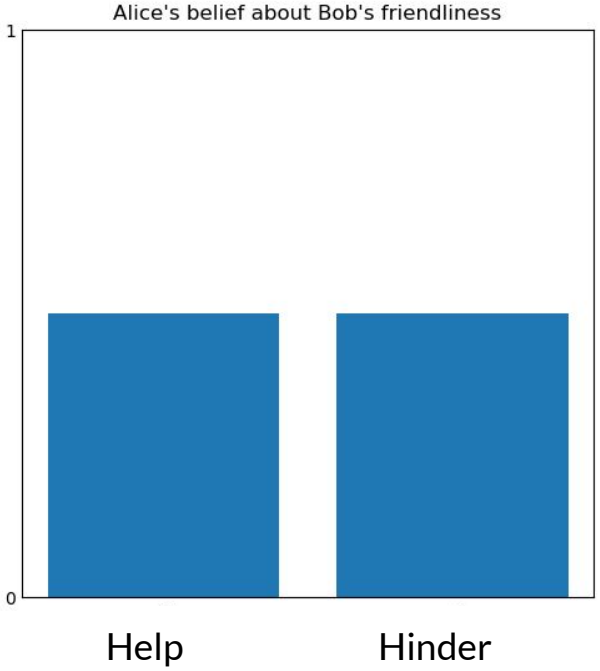


Experiment – Results from Bob Hindering Alice

Alice's Goals: and

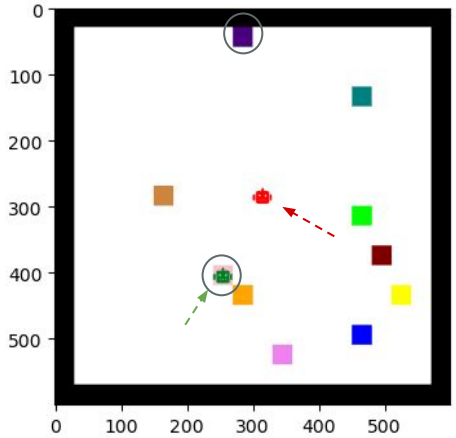


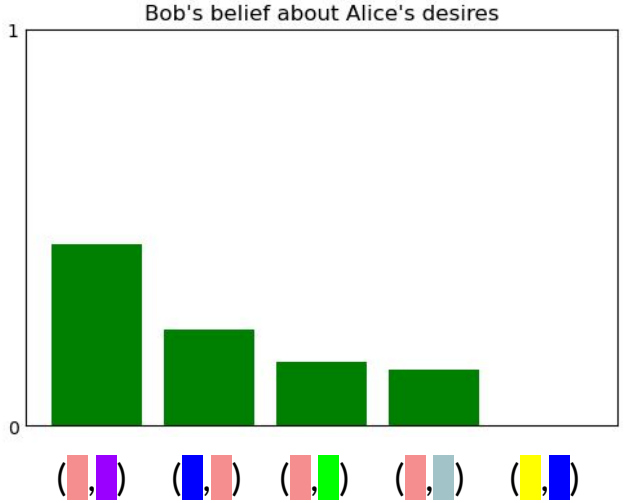


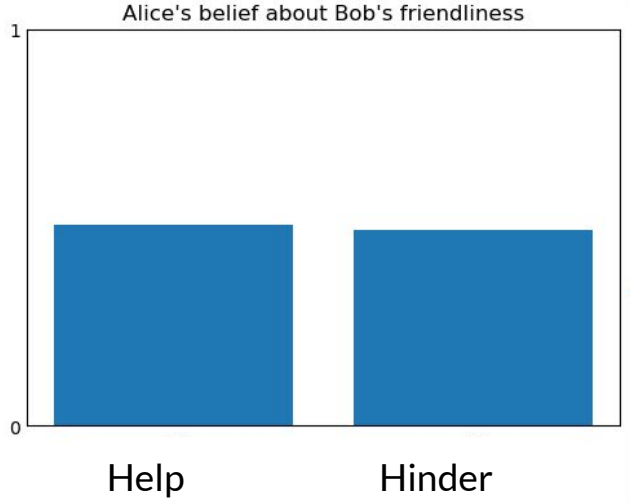


Experiment – Results from Bob Hindering Alice

Alice's Goals: and
Bob's Social Goal: Hinder

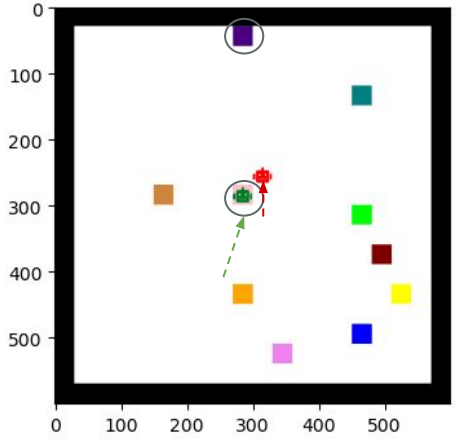


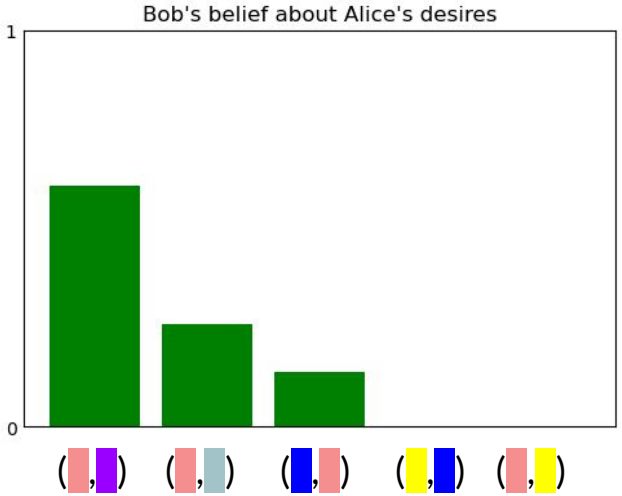


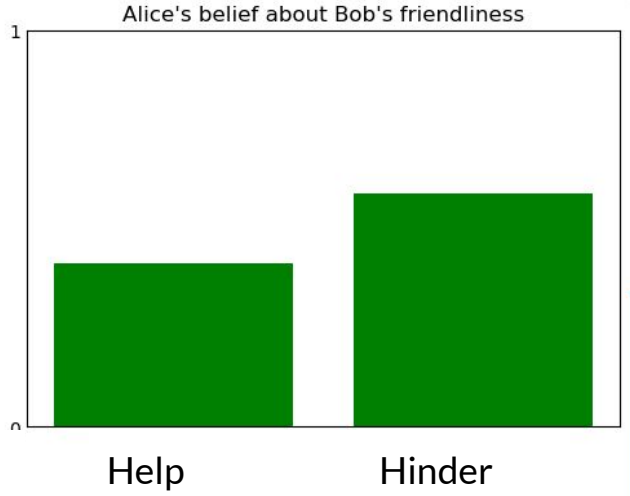


Experiment - Results from Bob Hindering Alice

Alice's Goals: ▮ and ▮
Bob's Social Goal: Hinder

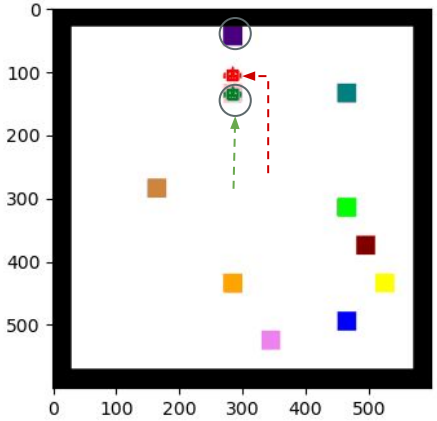


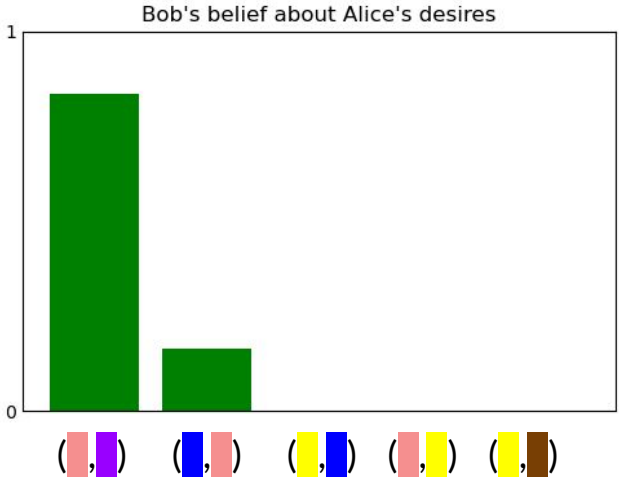


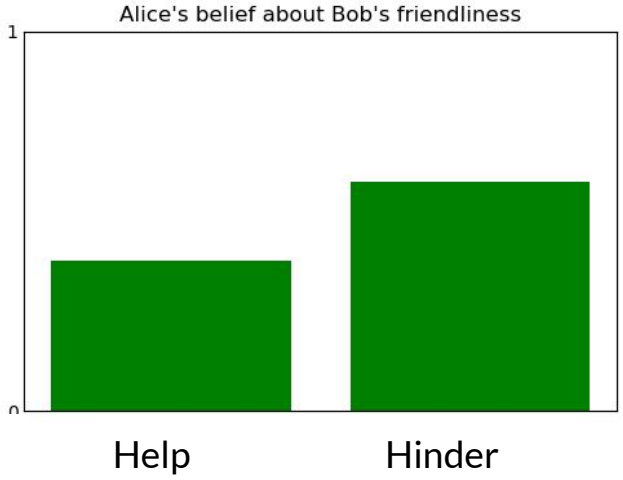


Experiment - Results from Bob Hindering Alice

Alice's Goals: ▮ and ▮
Bob's Social Goal: Hinder

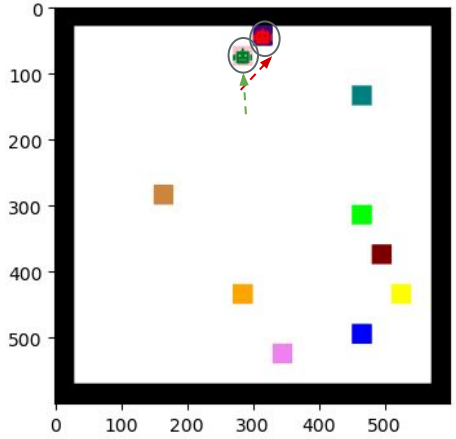


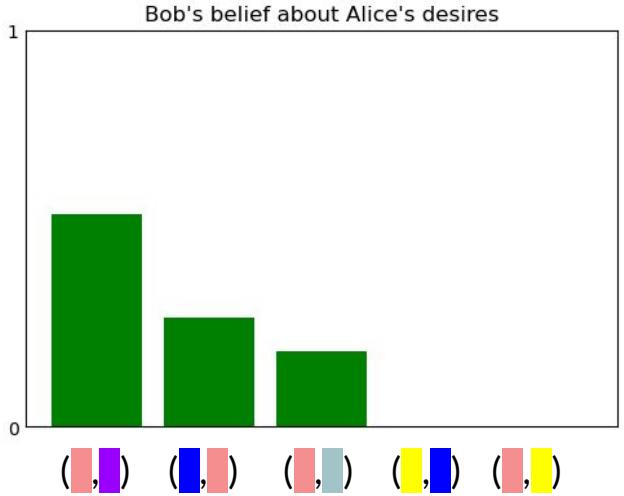


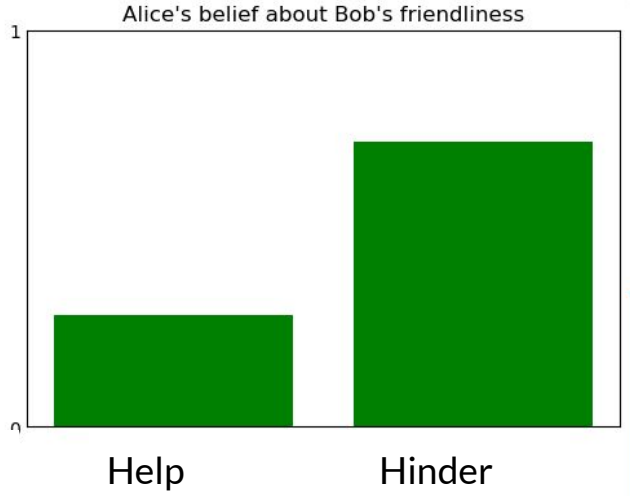


Experiment - Results from Bob Hindering Alice

Alice's Goals: ▮ and ▮
Bob's Social Goal: Hinder







Conclusion

- Neural amortized inference **drastically accelerates** nested social reasoning while preserving robustness and uncertainty estimation
- Current experiments focus on *goal* inference, but our framework extends to *physical state* inference

